

The dynamics of the system is given in (Analytic LQR Design for Spacecraft Control System Based on Quaternion Model) equations 2-6, which I copy below.

$$J\dot{\omega} = -\omega^\times(J\omega) + u \quad (1)$$

$$\omega^\times = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}. \quad (2)$$

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} f(q) & -q_3 & q_2 \\ q_3 & f(q) & -q_1 \\ -q_2 & q_1 & f(q) \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} = \frac{1}{2} \Omega \omega = g(q_1, q_2, q_3, \omega). \quad (3)$$

$$f(q) := q_0 = \sqrt{1 - q_1^2 - q_2^2 - q_3^2}. \quad (4)$$

The linearized system about $(q, \omega) = (\mathbf{q}_0, \boldsymbol{\omega}_0)$ is:

$$\delta \dot{\mathbf{q}} = \left. \frac{\partial g}{\partial \boldsymbol{\omega}} \right|_{(\mathbf{q}_0, \boldsymbol{\omega}_0)} \delta \boldsymbol{\omega} + \left. \frac{\partial g}{\partial \mathbf{q}} \right|_{(\mathbf{q}_0, \boldsymbol{\omega}_0)} \delta \mathbf{q}. \quad (5)$$

Where $\delta \mathbf{q} = \mathbf{q} - \mathbf{q}_0$, $\delta \boldsymbol{\omega} = \boldsymbol{\omega} - \boldsymbol{\omega}_0$, and :

$$\frac{\partial g}{\partial \boldsymbol{\omega}} = 1/2 \begin{bmatrix} f(q) & -q_3 & q_2 \\ q_3 & f(q) & -q_1 \\ -q_2 & q_1 & f(q) \end{bmatrix} \quad (6)$$

$$\frac{\partial g}{\partial \mathbf{q}} = 1/2 \begin{bmatrix} \frac{\partial f}{\partial q_1} \omega_1 & \frac{\partial f}{\partial q_2} \omega_1 + \omega_3 & \frac{\partial f}{\partial q_3} \omega_1 - \omega_2 \\ \frac{\partial f}{\partial q_1} \omega_2 - \omega_3 & \frac{\partial f}{\partial q_2} \omega_2 & \frac{\partial f}{\partial q_3} \omega_2 + \omega_1 \\ \frac{\partial f}{\partial q_1} \omega_3 + \omega_2 & \frac{\partial f}{\partial q_2} \omega_3 - \omega_1 & \frac{\partial f}{\partial q_3} \omega_3 \end{bmatrix} \quad (7)$$

The linearization of ω is:

$$\delta \dot{\boldsymbol{\omega}} = J^{-1} ((J\boldsymbol{\omega}_0)^\times - \boldsymbol{\omega}_0^\times J) \delta \boldsymbol{\omega} + J^{-1} \mathbf{u}. \quad (8)$$

Putting them together:

$$\begin{bmatrix} \delta \dot{\boldsymbol{\omega}} \\ \delta \dot{\mathbf{q}} \end{bmatrix} = \begin{bmatrix} J^{-1} ((J\boldsymbol{\omega}_0)^\times - \boldsymbol{\omega}_0^\times J) & 0 \\ \frac{\partial g}{\partial \boldsymbol{\omega}} & \frac{\partial g}{\partial \mathbf{q}} \end{bmatrix} \begin{bmatrix} \delta \boldsymbol{\omega} \\ \delta \mathbf{q} \end{bmatrix} + \begin{bmatrix} J^{-1} \\ 0 \end{bmatrix} u \quad (9)$$